Ankit Prajapati

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Education

Indian Institute of Technology, Madras May 2023 – May 2027
Bachelor of Science in Data Science

Indore Institute of Science and Technology, Indore Aug 2023 – June 2027
B.Tech in Computer Science Engineering

Delhi Public School, Sagar April 2021 – 2022
Higher Secondary (PCM + Informatics Practices)

Technical Skills

Languages: Python, C/C++, Java, JavaScript, SQL
Frameworks/Tools: Flask, Django, OpenCV, React, Node.js, PyTorch, TensorFlow
ML & AI: Machine Learning, Deep Learning, LLMs, RAG, LangChain,Prompt Engineering
Core CS: Data Structures and Algorithms, OOP, Operating Systems, System Design
Other: Linux, Git, FastAPI, Streamlit, Hugging Face, Firebase

Experience

Undergraduate Research Intern *Jul 2026 – Present*
Prof. Rahul H. Meshram, Dept. of EE, IIT Madras

- Implementing **Dynamic Programming** (Value Iteration, Policy Iteration) and **Q-Learning** algorithms to solve **Markov Decision Process (MDP)**-based sequential decision problems, aligned with the professor's research on stochastic sequential decision-making.
- Built a custom **grid-world simulation environment** in Python where an agent learns optimal navigation policies, benchmarking DP-based planning against Q-Learning's model-free approach to identify the most efficient path to the goal state.
- Tracking convergence behavior and policy performance using **TensorBoard**, with experiments version-controlled via **Git** on a **Linux**-based research workflow.

AI Engineer: Open Source Contributor *Jan 2025 – Apr 2025*
Klover.AI (Remote)

- Engineered **multi-agent AI workflows** combining research concepts with **production-ready systems** for automated decision-making and fault recovery.
- Built **modular, reusable agent components** improving reasoning consistency, reliability, and maintainability.
- Collaborated with distributed contributors to translate experimental ideas into **deployable AI pipelines**.

Projects

Backend Pilot: AI-Powered Backend Code Generator **Project Link**

- Built LLM-powered code generation system using Groq APIs to generate backend services (APIs, folder structure, tests) from natural language prompts.
- Designed prompt pipelines and structured outputs to ensure consistent, production-ready code generation.

SERA: Smart Power Line Fault Detection & Response – SIH 2025 Winner | PS: 25063 | KSEBL **Project Link**

- Led development of an **IoT + Edge AI solution** to detect **High Impedance Faults (HIF)** in low-voltage distribution lines missed by traditional protection.
- Designed a **lightweight LSTM-based edge model** for real-time current waveform analysis with automatic phase isolation and remote monitoring.
- Validated fault detection logic through simulated fault scenarios and edge-level testing.

SignEcho – Real-Time Sign Language Translator **Project Link**
Winner, Apratim 2025 IIST Indore – Internal Hackathon (200+ teams, 30-hour)

- Real-time sign-language translator using **CNN + MediaPipe** with **79.6% accuracy**, converting hand gestures into instant speech and text; deployed as a user-tested web app (OpenCV, PyTorch, TTS APIs).

EduVote – Secure Digital Voting System

- Full-stack digital voting system with secure authentication, role-based access, and real-time election analytics for transparent student elections (React, Node.js, Express, MongoDB, Firebase Auth).

Saathi – Smart Medicine Dispenser **Project Link**
IISF 2024 – Science and Technology Hackathon at IIT Guwahati (30-hour)

- AI-powered health monitoring system with medicine reminders, heartbeat and fall detection, and caregiver alerts; contributed backend logic and sensor integration using ESP32, Arduino, TensorFlow, Flutter, and Firebase.

Honors & Achievements

- **Smart India Hackathon (SIH) 2025 Winner**: Winner among top teams nationally | PS: 25063 | **View Certificate**
- **Apratim Srajan Hackathon 2025 Winner (IIST Indore)**: Winner among 200 teams. | **View Certificate**
- **Hacksagon 2025 (ABV-IIITM IEEE) Finalist**: Competed nationally at ABV-IIITM Gwalior. | **View Certificate**
- **S&T IISF 2024 Finalist**: Indian International Science Festival 2024 at **IIT Guwahati**. | **View Certificate**
- **Runner-up, Cognizance Watch Tower Hackathon 2023 (IIT Roorkee)**: Top-performing team in hardware, AI integration challenge. | **View Certificate**